

Acoustic Biodiversity Mapping

A global location-level biodiversity metric from Xeno-canto recordings

Generated 2026-07-08

OBJECTIVE

Derive a single metric that classifies a geographic location as biodiversity 'good' or 'bad', using a global archive of wildlife sound recordings and their metadata.

DATA

- 207,601 recordings across 5 continents (Xeno-canto), each with 42 acoustic indices computed by a SLURM batch pipeline (compute_indice.py over Butterworth-filtered audio).
- Metadata flattened from 274,296 API records (species, coordinates, quality, device).
- Acoustic scores joined to metadata on recording id (>99.99% match).

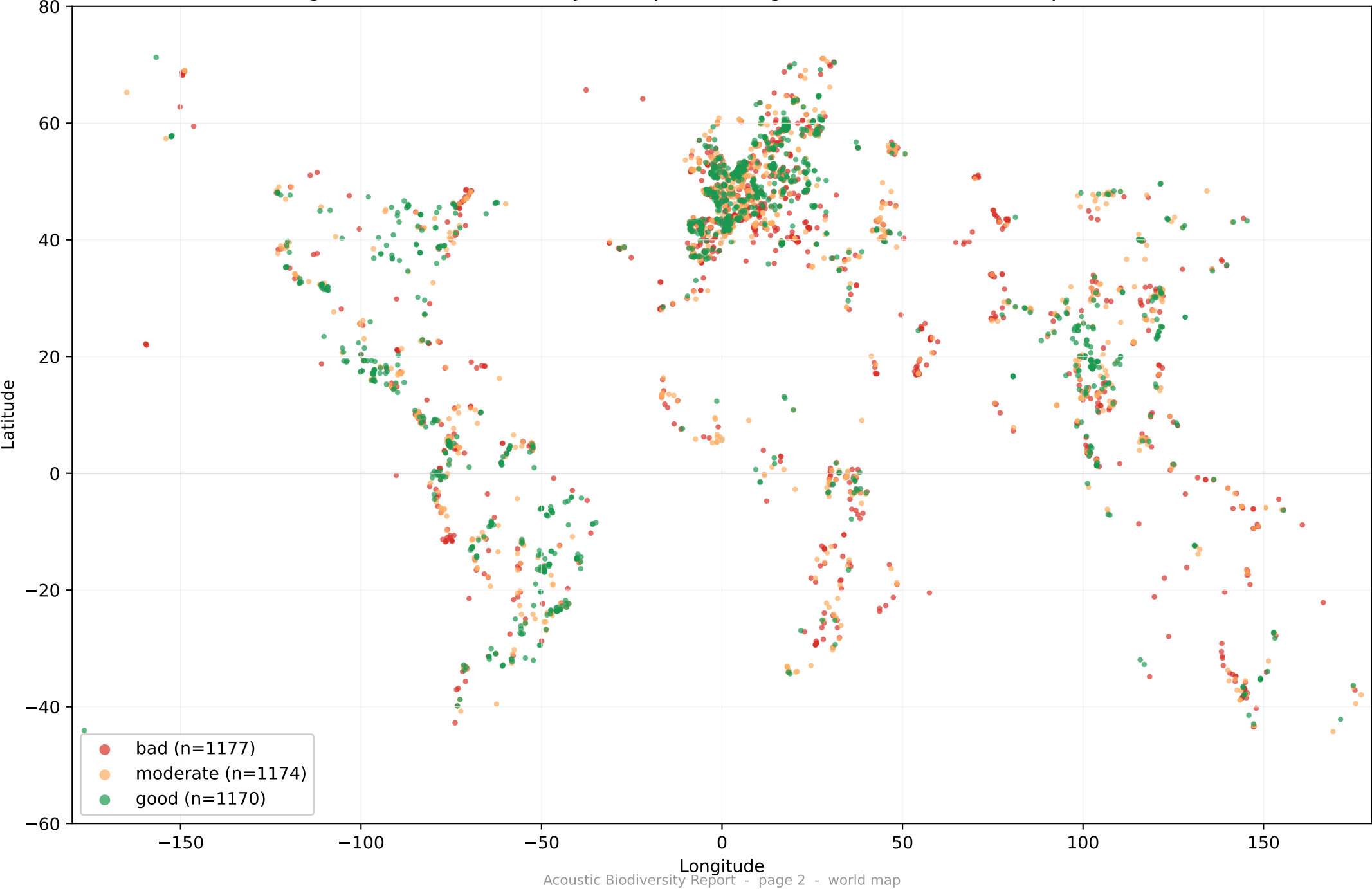
METHOD

1. Aggregate all recordings into 0.1-degree (~11 km) grid cells -> 16,212 cells.
2. Per cell, compute species richness from recorded species (genus+species and the 'also' co-occurring-species field), corrected for sampling effort by Hurlbert rarefaction to 10 recordings (S_{rare10}).
3. Test whether the acoustic indices predict richness (validation against ground truth).
4. Label each well-sampled cell good / moderate / bad relative to its 10-degree latitude band (region-relative tertiles).

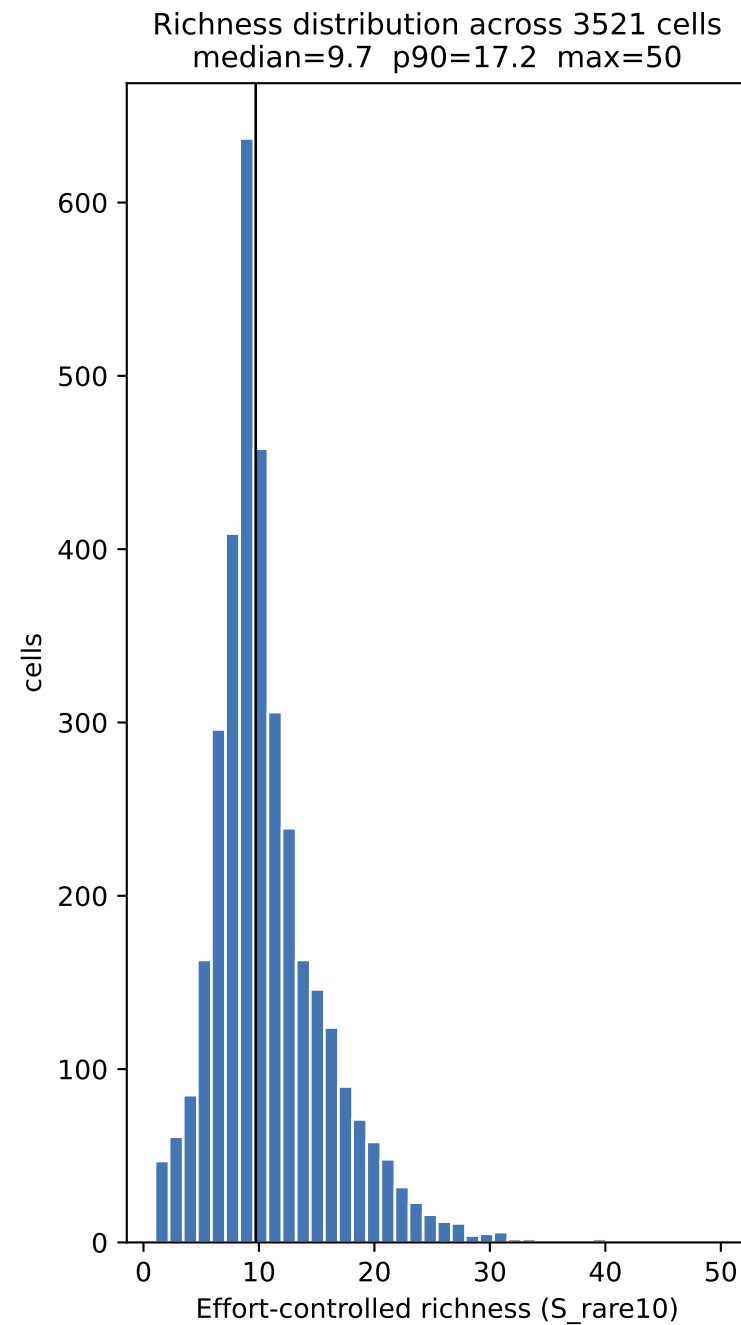
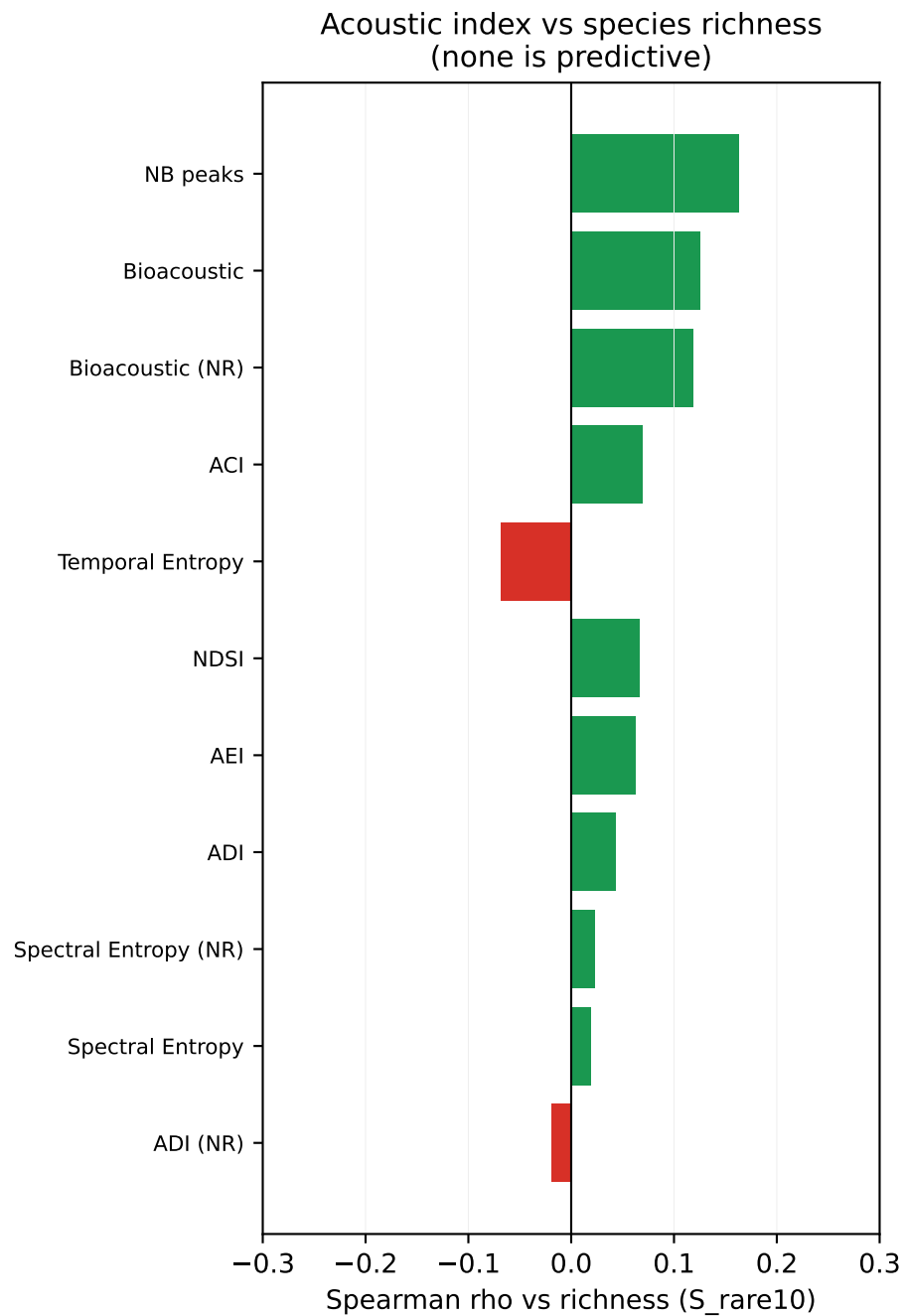
HEADLINE RESULT

- 3,521 cells had enough recordings (≥ 10) to score confidently.
- The acoustic indices did NOT predict species richness (best |Spearman rho| ~ 0.16). The defensible biodiversity metric is therefore effort-controlled species richness, not any acoustic index -- the indices are unreliable on targeted (non-soundscape) recordings that dominate the archive.
- Final labels: 1170 good, 1174 moderate, 1177 bad.

Region-relative biodiversity label per 0.1-degree cell (3,521 well-sampled cells)



Validation: does the audio carry the biodiversity signal?



The Metric & Results

METRIC DEFINITION

location = 0.1 deg x 0.1 deg grid cell (~11 km)
biodiversity value = S_{rare10} , the expected number of distinct recorded species in 10 recordings (Hurlbert sample-based rarefaction; controls for recording effort)
label = tertile of S_{rare10} WITHIN the cell's 10-deg latitude band: good = top third, bad = bottom third
scope = cells with ≥ 10 recordings (3,521 of 16,212 cells)

GLOBAL VALUE RANGE (S_{rare10})

min 1.0 p25 7.9 median 9.7 p75 12.9 p90 17.2 max 50

Median richness by continent (well-sampled cells)

Continent	Median richness	Scored cells
america	10.8	802
europa	9.7	1767
asia	9.2	608
africa	9.1	233
australia	8.6	111

KEY LIMITATIONS (must accompany any use of this metric)

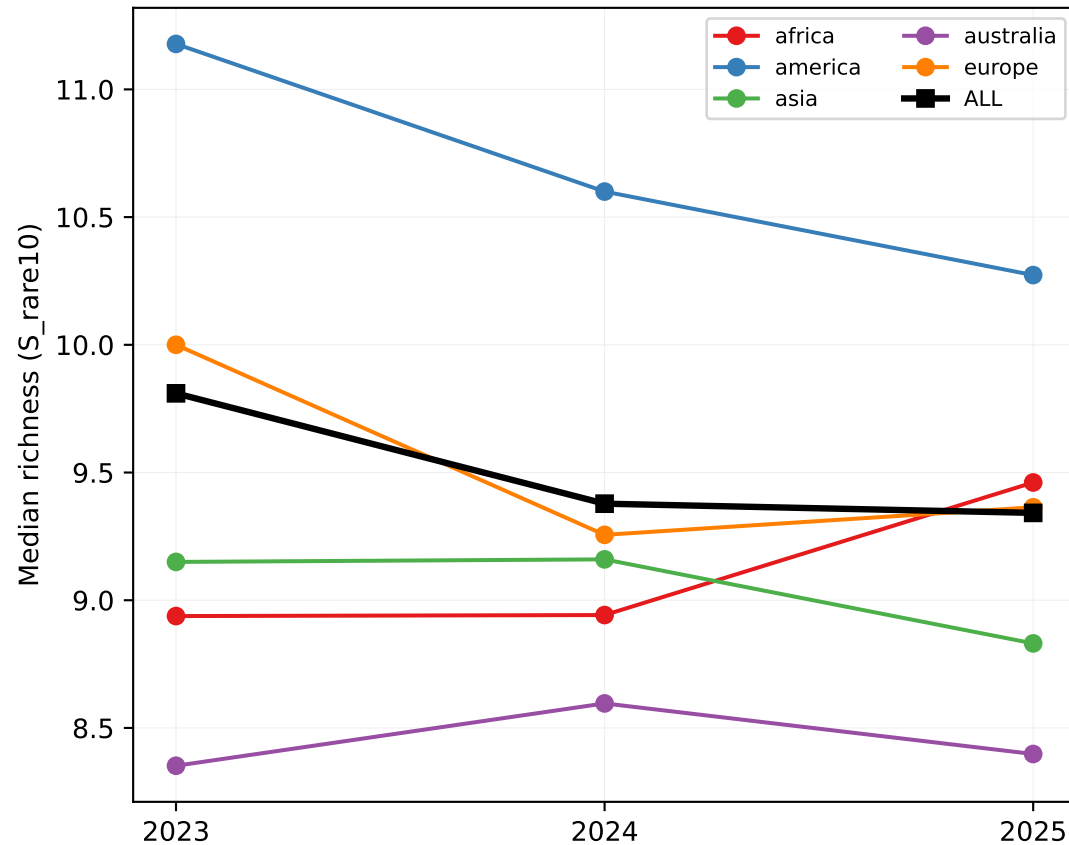
1. Acoustic indices are non-predictive here. ACI/ADI/NDISI etc. were built for passive soundscape monitoring; 69% of archive clips are single-target recordings (median 24 s), so per-clip indices do not reflect site diversity.
2. 'Richness' = RECORDED species, not true species. It reflects recordist effort and interest. Rarefaction controls sample SIZE but not observer bias.
3. Weak latitude gradient. Effort-controlled richness is nearly flat across latitude (thresholds ~10-13 everywhere) -- evidence the signal is effort-driven, not the true tropical biodiversity peak. Region-relative labelling is used so scores mean 'rich for its region'.
4. Coverage. Only 22% of cells are confidently scored; the rest are 'insufficient_data' (< 10 recordings).

OUTPUT FILES

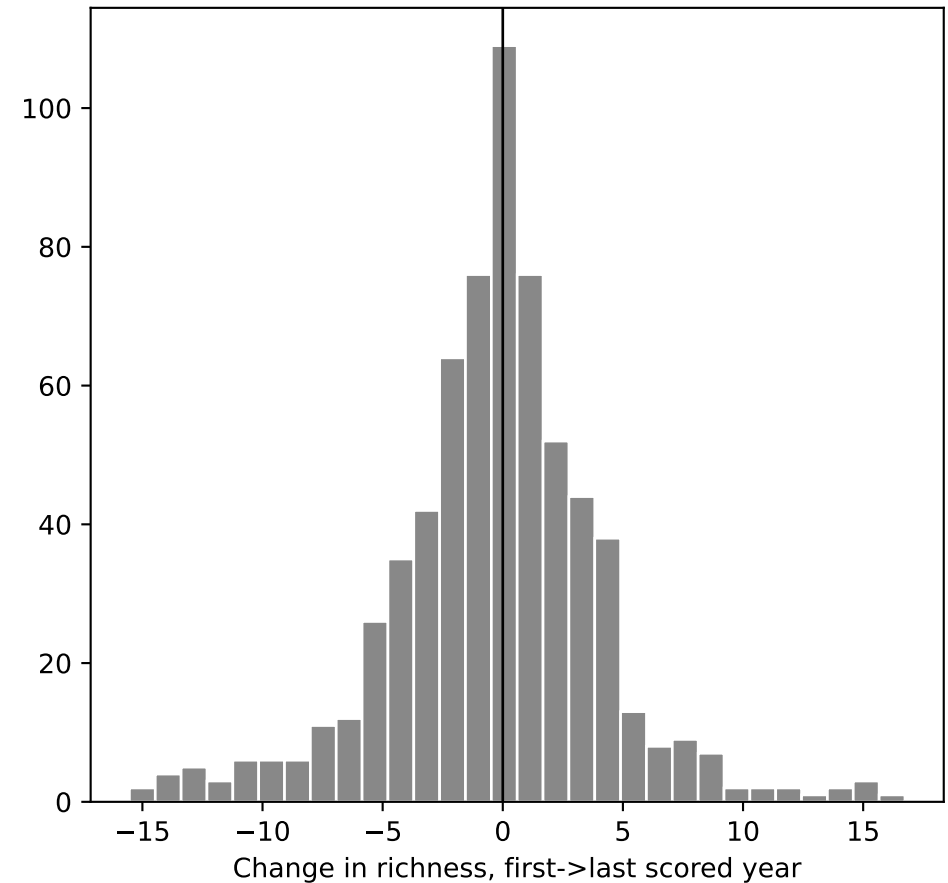
grid_cells.csv	- per-cell features + richness (all 16,212 cells)
grid_cells_labeled_regional.csv	- region-relative good/moderate/bad labels
biodiversity_map.csv	- lat, lon, label, code -> ready for GIS/folium

Year-by-year (2023-2025): temporal slices of the metric

Recorded-richness trend by continent



Per-cell change (667 cells tracked)
up=266 down=296 flat=105



COVERAGE: 3,956 scored cell-years; 667 cells scored in ≥ 2 years, 212 in all three.

CAUTION: a 3-year window cannot show real biodiversity change. These movements reflect WHICH cells were recorded and by WHOM each year (effort + observer turnover), not ecological gain or loss. The global dip (9.8 $\rightarrow$ 9.3) tracks recording effort, not nature. Use year slices to study sampling coverage over time -- not as a biodiversity time series.

OUTPUTS: grid_cells_yearly.csv (per cell-year), cell_change.csv (per tracked cell).